SEATRACE AI — OFFICIAL MARITIME INCIDENT DOSSIER

CONFIDENTIAL INVESTIGATION DOSSIER • INCIDENT IN-2026-0881 • GENERATED AT 2026-10-23 12:16:52 UTC

1. Incident Summary & Geographic Coordinates

Incident ID	IN-2026-0881	**Classification**	CRITICAL
Title	Mumbai High Offshore Tanker Sheen	**Status**	RECONSTRUCTING
Maritime Region	Arabian Sea (West Coast)	**Coordinates**	19.3421°N, 71.8542°E
Detection Time	2026-10-22T06:14:00Z	**Estimated Volume**	340.5 m³
Sensor / Mission	Sentinel-1A SAR (IW Mode)	**Spill Type**	Heavy Fuel Oil / Crude

2. Satellite Detection & Look-Alike Validation

Analyzed scene **S1A_IW_GRDH_1SDV_20261022T061400_051280_062F1A** captured by Sentinel-1A C-SAR (VV Polarization). Estimated slick footprint area is **14.85 km²** with perimeter 18.40 km. Algorithmic confidence score is **94.0%**.

Validation Decision: **SUPPORTED** by reviewer Dr. Aris Thorne (Lead Satellite Oceanographer). Surface wind speed: 6.8 kts. Biogenic surfactant risk: LOW. Proximity to charted natural seep: 142.5 km. Notes: *Confirmed petroleum hydrocarbon characteristics. Damping and wind velocity strongly rule out natural surfactant look-alikes.*

3. Environmental Origin Reconstruction

Lagrangian backward trajectory simulation executed using **500 particles** over a **24-hour window** with windage leeway 3.2%. Estimated release origin centroid is located at **19.1840°N, 71.7020°E** (spatial uncertainty ±6.4 km). Forcing models: INCOIS Ocean Current (HYCOM 1/12 deg) + NOAA GFS 10m Wind.

4. AIS Vessel Trajectory Correlation & Attribution

Rank	MMSI	Vessel Name	Flag	Type	Dist (km)	Time Δ	Score	Lead Status
#1	419001842	MT NEELKANTH GLORY	India	Crude Oil Tanker	1.8	+0.5h	94.6%	CRITICAL_LEAD
#2	352002781	PACIFIC TRADER V	Panama	Chemical/Product Tanker	5.6	-1.8h	72.4%	ELEVATED
#3	636019441	EVER PROSPER	Liberia	Container Ship	14.2	-3.5h	38.5%	LOW_COMPATIBILITY

5. Cryptographic Evidence Chain of Custody

Artifact Type	Title	SHA-256 Checksum Hash	Verified
RAW_SAR_SCENE	Sentinel-1A IW L1 GRD Scene (Safe Format)	79c8136830800c03...a748f760	VERIFIED
DETECTION_MASK	Binary Segmentation Slick Polygon (GeoJSON)	cb6e3df1cbff8571...26d00bbb	VERIFIED
AIS_CORRIDOR_EXTRACT	Raw AIS Transponder Stream - Arabian Sea Box A4	48c1b8194ca3f60e...f6fabd54	VERIFIED
DRIFT_SIMULATION_LOG	Lagrangian Particle Tracking Trajectory Ledger	fb8ce76484c74f36...0ae89914	VERIFIED

LEGAL NOTICE: This investigative dossier contains probabilistic modeling, satellite remote sensing interpretations, and AIS track correlation. Vessel rankings constitute candidate investigative leads for competent maritime enforcement authorities and do not constitute self-executing legal guilt without field

corroboration.